

Modbench

User manual

A Modbus client for Windows, Linux and macOS: manual requests, device maps, charts, a simulator and tools to find and check devices.

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github.com/nobrega8/modbench

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CHAPTER 1

Introduction

Modbench is a Modbus client: it sends requests to Modbus devices (meters, inverters, drives, PLCs, controllers) and shows what they answer. It reads and writes registers by hand, or by name with a *device map*, charts values over time, shows every frame on the wire, and simulates devices when there is no hardware at hand.

It runs on Windows, Linux and macOS, speaks Modbus over serial lines (RTU and ASCII) and over the network (TCP, UDP, and RTU frames over TCP or UDP), and keeps everything on your computer.

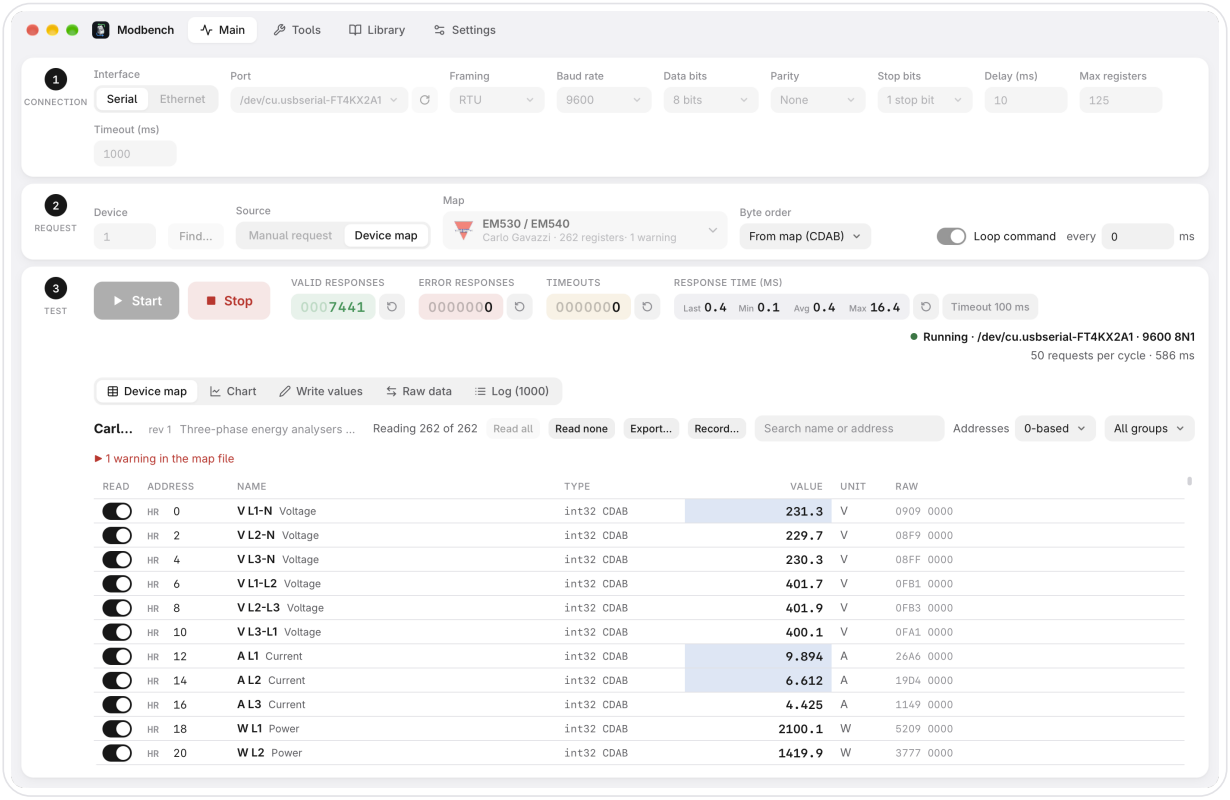


Figure 1. The Main page with a device map open.

1.1 What you can do with it

- **Send any request**: choose a function code, an address and a quantity, press Start, and see the values, once or in a loop.

- **Read a device by name**: open a map (`.mbmap.json`) and every register shows with its name, unit, scaling and state labels. Over 400 maps can be downloaded from the shared library.
- **Check that the map fits**: reference values catch a wrong address, base or byte order before you trust the other values.
- **Write safely**: typed controls, the map's limits, an optional confirmation, and a read-only mode that blocks every write.
- **See what happens**: charts, the raw bytes of each frame decoded field by field, a log of every request, and statistics of failures by cause.
- **Find and identify devices**: scan unit IDs and serial settings, find which addresses answer, ask a device who it is.
- **Simulate devices**: several devices shaped after maps, with values that move on their own, to try maps and other programs.

1.2 A few Modbus words

If you work with Modbus every day, skip this section.

Client and server
The client (Modbench) asks; the server (the device, also called slave) answers. A device never speaks first.

Unit ID
The address of a device on a serial line, or behind a gateway: `1` to `247`. Over TCP to a single device it is often ignored, but must still be sent (usually `1` or `255`).

Areas
Each device has four tables of data:

Area	Size	Read / write
Coils	1 bit	read (FC01), write (FC05, FC15)
Discrete inputs	1 bit	read only (FC02)
Input registers	16 bits	read only (FC04)
Holding registers	16 bits	read (FC03), write (FC06, FC16)

Address
The number of a coil or register in its area, from `0` to `65535` on the wire. Manuals often count from 1, or write the area in front (`40001` is holding register 1, that is 0 on the wire). Modbench shows addresses the way you choose (Section 3.3).

Byte and word order
A register holds two bytes; 32- and 64-bit values take two or four registers. Devices disagree on which byte and which register comes first, so the same bytes can read as very different numbers. Maps say which order a device uses; the Data decoder (Section 10.3) helps find it.

Exception
An answer that says the request failed, with a code: `02` for an address the device does not have, `03` for a value it refuses, and so on (Appendix A.4).

1.3 The window

Four pages sit in the title bar. Click one to show it.

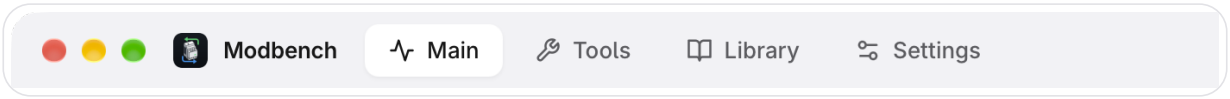


Figure 2. The title bar and its four pages.

Main
Connect to a device and test it, in three steps: *Connection*, *Request* and *Test* (Chapter 3).

Tools
The simulator, and tools to identify devices, find addresses, decode data and frames, play back recordings, see statistics and set alarms (Chapters 9 and 10).

Library
The device maps on this computer, and the shared online library (Chapter 8).

Settings
Theme, text size, safety, what Modbench remembers, updates (Chapter 11).

Right-click menus

Most lists and values have a right-click menu with what can be done there: copy a value or an address, add a register to the chart, write a value, export, remove. Text fields have **Cut**, **Copy**, **Paste** and **Select all**.

Light and dark

Modbench follows the system theme, or the one chosen in **Settings › Appearance**. The screenshots in this manual use the light theme.

CHAPTER 2

Installation and updates

Installers are on the releases page: `github.com/nobrega8/modbench/releases`. Each release lists what changed.

2.1 Windows

Download `Modbench_x.y.z_x64-setup.exe` (recommended) or the `.msi`, and run it. The installer is about 2 MB: Modbench uses the WebView2 that comes with Windows.

2.2 macOS

Download `Modbench_x.y.z_universal.dmg`: one app for Apple Silicon and Intel Macs. Open it and drag Modbench to **Applications**.

> First start
>
> Modbench is not signed with an Apple Developer ID, so macOS asks before the first start. Right-click the app in **Applications**, choose **Open**, then **Open** again. Later starts need nothing.

2.3 Linux

- **AppImage**: download it, make it executable (`chmod +x Modbench*.AppImage`) and run it.
- **Debian, Ubuntu and derivatives**: install the `.deb`, or add the APT repository so that `apt upgrade` brings new versions:

```
curl -fsSL https://nobrega8.github.io/modbench/key.asc | sudo gpg --dearmor -o /usr/
    share/keyrings/modbench.gpg
echo "deb [arch=amd64 signed-by=/usr/share/keyrings/modbench.gpg] https://nobrega8.
    github.io/modbench stable main" | sudo tee /etc/apt/sources.list.d/modbench.list
sudo apt update && sudo apt install modbench
```

> Serial ports on Linux
>
> To open serial ports without root, add your user to the `dialout` group (`sudo usermod -aG dialout $USER`) and log in again.

2.4 Updates

Modbench checks for a new version when it starts (this can be turned off in **Settings › Updates**). When there is one, a banner says what changed:

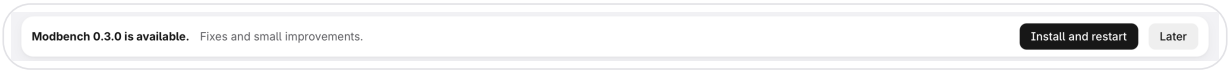


Figure 3. The update banner.

Install and restart downloads the update, checks its signature, installs it and starts Modbench again. Later hides the banner until the next start. **Settings › Updates** can also check by hand.

Installs from the APT repository update with the system (`sudo apt upgrade`) instead.

2.5 Files Modbench opens

The installers register two file types, each with its own icon. A double-click opens them in Modbench, also when it is already running.

File	What it is
`.mbmap.json`	A device map (Chapter 4). Opens on the Main page.
`.mbmap`	A device map in the older version 1 format; Modbench can convert it.
`.mbsim`	A simulator setup (Section 9.6). Opens in the simulator.

> **Note**
> Windows tells file types apart by their last extension only, so `.mbmap.json` files keep the icon of JSON files there. On Linux they get the map icon.

CHAPTER 3

The Main page

The Main page tests a device in three steps, from top to bottom: *Connection* (how to reach it), *Request* (what to ask) and *Test* (send, and see the answers).

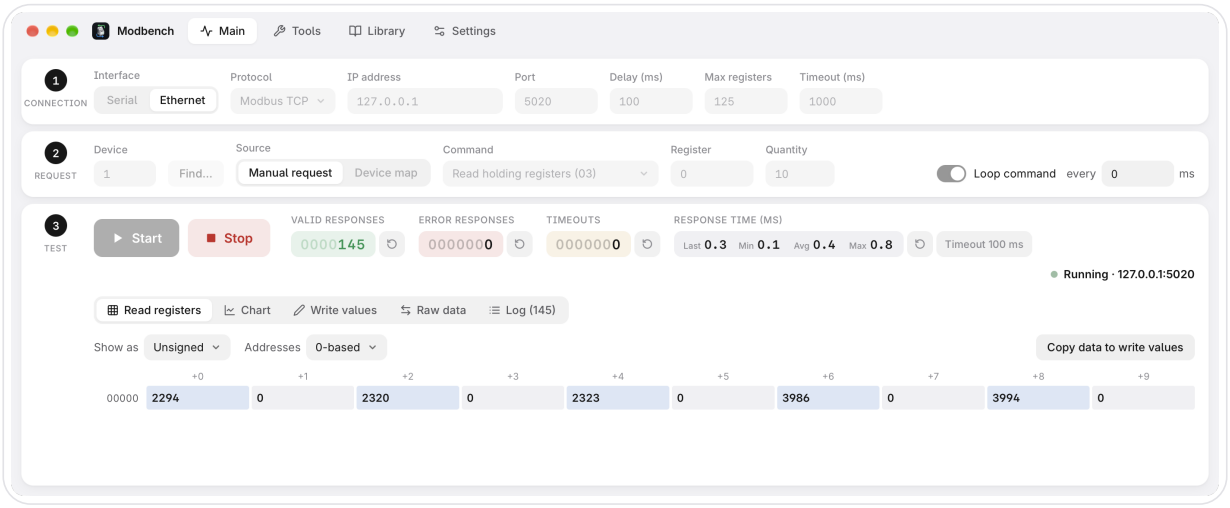


Figure 4. The Main page: Connection, Request and Test.

3.1 Step 1: Connection

Interface

Choose how the device is reached:

Interface	Use it for
Serial, RTU	RS-485 or RS-232 devices: the usual Modbus on a serial line.
Serial, ASCII	Older devices that send frames as text, checked with an LRC.
Modbus TCP	Devices and gateways on Ethernet (port 502).
Modbus UDP	Devices that answer Modbus TCP frames over UDP.
RTU over TCP	Serial-to-Ethernet converters that pass RTU frames as they are.
RTU over UDP	The same, over UDP.

SERIAL

Serial port lists the ports of this computer (↻ looks again after plugging an adapter in). Set **Baud rate**, **Data bits**, **Parity** and **Stop bits** as the device is configured: every device on the line must use the same.

NETWORK

IP address and **Port**. The port is 502 unless the device or the converter uses another.

Device

Device (unit ID) is the device's address: 1–247 on a serial line or behind a gateway. **Timeout (ms)** is how long to wait for an answer before counting a timeout; **Delay (ms)** is the pause between two requests, which slow devices and some converters need.

> **Not sure of the settings?**
> **Find...** scans for unit IDs that answer, or, on a serial line, for the baud rate, parity and stop bits of a device (Section 3.5). The Statistics tool suggests a timeout from the measured response times (Section 10.6).

3.2 Step 2: Request

Source chooses between a *manual request* and a *device map*.

Manual request

Command
The function code: read coils (FC01), discrete inputs (FC02), holding registers (FC03) or input registers (FC04), or one of the writes (FC05, FC06, FC15, FC16).

Start address
The first address, in the notation chosen in Settings (below).

Quantity
How many coils or registers. Modbus allows up to 125 registers or 2000 bits per read.

Device map

With **Source** set to a map, **Map** chooses one of the maps of the library, an open file, or **From a file...**. Modbench then reads every register of the map by name (Chapter 4). The map's connection defaults (unit ID, baud rate, parity, port) fill step 1 when the map is picked; you can still change them.

Byte order

Byte order reads every value of a map with another byte and word order than the map says. It is meant for checking: if the values only make sense with another order, the map, or the device's configuration, is wrong.

Loop

With **Loop command** on, the request repeats until **Stop**. The time next to it is from the start of one cycle to the start of the next; `0` repeats as soon as **Delay** allows. It can be changed while the test runs. Without Loop, **Start** sends the request once.

3.3 Address notation

Manuals write addresses in many ways. The address menu on the **Device map**, **Read registers** and **Log** tabs chooses how Modbench shows them:

Notation	Holding register 100	Notes
0-based	`100`	The address on the wire.
1-based	`101`	As many manuals number registers.
Hex	`0x0064`	As in protocol documents.
Modicon	`400101`	Area digit in front (`0`, `1`, `3`, `4`), 1-based.
IEC 61131-3	`%MW100`	As PLC programs write them.

Address fields also take the Modicon and IEC forms as typed, and switch the command to their area: typing `300010` reads input register 9.

3.4 Step 3: Test

Start sends the request (once, or in a loop); **Stop** ends it. The counters show **Valid responses**, **Error responses** (exceptions) and **Timeouts**; the button next to each counter sets it back to zero.

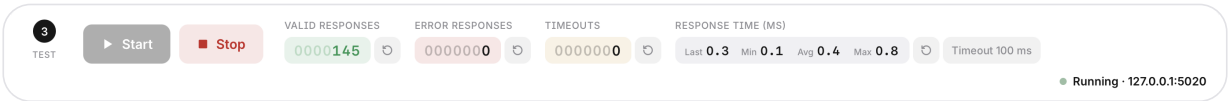


Figure 5. Start, Stop and the counters.

The tabs below show the results:

Device map or **Read registers**
The values: by register name with a map (Chapter 4), as a grid for a manual read (Chapter 5).

Chart
Values over time (Chapter 6).

Write values
Write coils and registers (Section 5.2).

Raw data
The bytes of each frame, decoded (Section 7.1).

Log
Every request with its result and time (Section 7.2).

When the link drops

With **Reconnect when the connection is lost** on (in Settings), a test that loses its link (a cable pulled, a gateway restarted) keeps trying to reconnect, and goes on where it stopped.

3.5 Finding devices

Find... in step 1 opens a scan:

Unit IDs
Asks every unit ID in a range (1–247 by default) and lists the ones that answer. Use a short **Wait (ms)** to go faster on a quiet line.

Serial settings
For a serial device whose settings are unknown: tries the chosen baud rates, parities and stop bits, most common first, until the device answers.

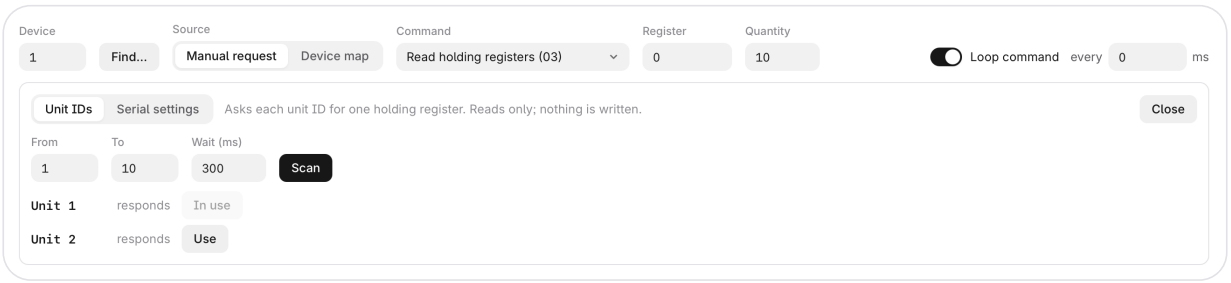


Figure 6. Finding unit IDs and serial settings.

Use next to a unit ID that answered puts it in step 1.

CHAPTER 4

Device maps

A device map describes the registers of a device model: their names, areas, addresses, types, byte and word order, scaling, units and state labels, plus the device's connection defaults. With a map, Modbench reads a device by name, as its manual describes it, instead of as a list of numbers.

Maps are `.mbmap.json` files in an open format. Over 400 of them, for meters, inverters, drives and controllers, are shared in the *mbmap-library* and can be downloaded from the Library page (Chapter 8).

4.1 Opening a map

- In step 2 of the Main page, set **Source** to the map, and pick it in **Map**: the maps of your library are listed there.
- **From a file...** opens any `.mbmap.json` or `.mbmap`.
- Double-click a map file: it opens in Modbench.
- From the Library page: right-click a map and choose **Use**.

Several maps can be open at once; step 2 switches between them. The map's connection defaults (unit ID, port, baud rate, parity) fill step 1 when it is opened.

> Older maps
>
> A `.mbmap` file in the version 1 format opens with a warning, and Modbench offers to convert it: the converted copy is saved next to it as `.mbmap.json`, and the original is kept.

4.2 The Device map tab

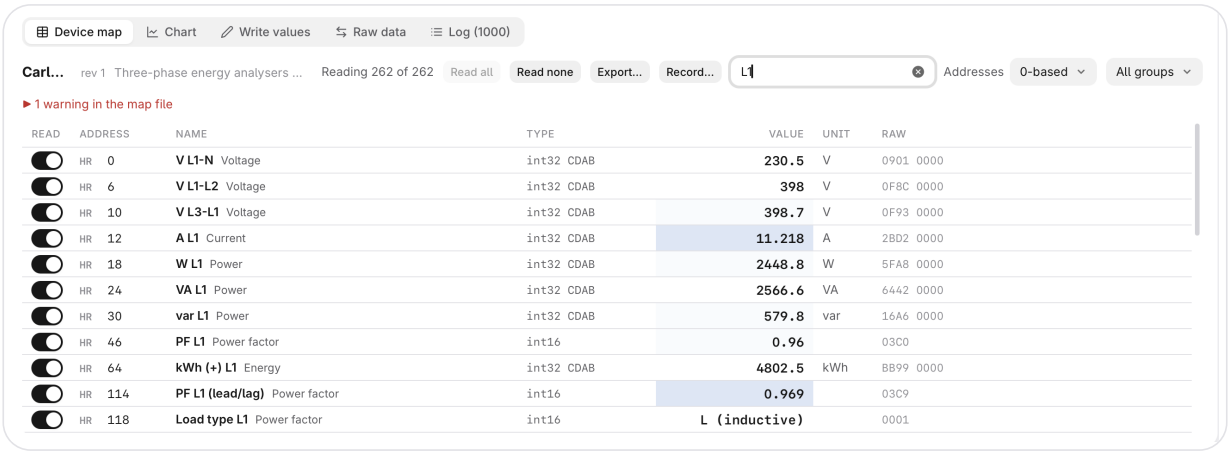


Figure 7. The Device map tab during a test.

Every register of the map is a row: its name, address, value with its unit, and its group. Values update with each cycle of the test.

Choosing what to read
Untick a register to stop reading it; Read all and Read none change them all. Fewer registers make each cycle shorter. Modbench joins neighbouring registers into as few requests as the device allows.

Search
Search name or address filters the list.

State labels
Registers that hold a state (`0` Stopped, `1` Running...) show its label; reserved values the map declares (such as `0xFFFF` for "not implemented") show their meaning instead of a number.

Bits and bit fields
Registers that pack several flags show each bit or field by its own name.

Changed values
A value that changes stays highlighted for a while (**Settings** › **Values**), so changes stand out in a long list.

Right-click
Copy a value, a name or an address, add the register to the chart, or write it.

Registers the map is missing

Add a register the map is missing... adds a register on this computer only: its address, type and name. It is read with the others and marked as added; it is not written into the map file.

4.3 Is it the right map?

A wrong map, base address or byte order gives values that look like numbers but mean nothing. Modbench checks in two ways:

Reference values
A map can give the value a healthy device holds in some registers (a model code, a fixed protocol identifier). Modbench reads them and shows *match*, *mismatch* or *failed* (no answer). A mismatch points to a wrong map or settings.

Models and firmware
When the map says which models and firmware versions it fits, and has registers that identify the device, Modbench compares them with the connected device. *Device recognised* confirms it; otherwise a banner says the device does not match the map. Once the firmware is known, registers that do not exist in that version are left out.

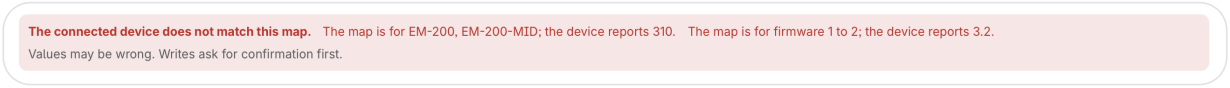


Figure 8. A device that does not match the map.

> **Careful**
> After a mismatch, writes ask for confirmation even when confirmation is off.

4.4 Map status

Maps from the library carry a status, shown as an icon in the library and in the map picker:

Status	Meaning
Verified	Checked against a real device.
Unverified	Written from the manufacturer's documentation, not yet checked on a device.
Unofficial	Not from the manufacturer's documentation, for example reverse-engineered.

4.5 Byte order override

Byte order in step 2 reads every value with another byte and word order than the map says: *as the map says*, or one of the four orders. Use it to check a map whose values look wrong; it is remembered per device.

4.6 Exporting and recording

Export...
Saves the values shown, in the order shown, to a CSV file.

Record...
Appends every cycle of the test to a CSV file until recording stops: one row per cycle, one column per register. The Trend player plays such a file back (Section 10.5).

The CSV separator is set in **Settings** (comma or semicolon, for spreadsheets in languages that use a decimal comma).

4.7 What is remembered per device

For each device (map, connection and unit ID), Modbench remembers which registers were switched off, the byte order override and Loop. They come back when you return to that device; restoring them never writes and never starts a test. **Settings** › **Connection** turns this off, and forgets what was saved.

CHAPTER 5

Reading and writing

5.1 The Read registers tab

With a manual request, the values come back as a grid: one cell per coil or register, with its address.



Figure 9. Holding registers read as 32-bit values.

Show as
Shows the registers as unsigned or signed 16-bit numbers, hexadecimal, binary, ASCII text, 32-bit signed or unsigned integers, or 32-bit floats, in the byte order chosen next to it. When the right type is not known, the Data decoder shows them all at once, 64-bit types included (Section 10.3).

Bits
Shows the 16 bits of each register. A bit of a holding register can be switched on its own: Modbench writes the register with only that bit changed.

Addresses
The address menu chooses the notation (Section 3.3).

Right-click
Copy a value, an address or all values, add the address to the chart, or **Write here…** and **Write all…**.

5.2 Writing values

The `Write values` tab writes coils and registers.

Without a map

Choose the function (single or multiple coils, single or multiple registers), the start address and the values, then **Write**. `Copy data to write values` fills the list with the values last read, to change one and write them back.

With a map

The map’s writable registers are listed by name, each with a control for its type: a number in engineering units, a list of the map’s labels for state registers, a switch for bits. Modbench applies the inverse of the scaling, checks the type’s range and the map’s limits, and only then writes.

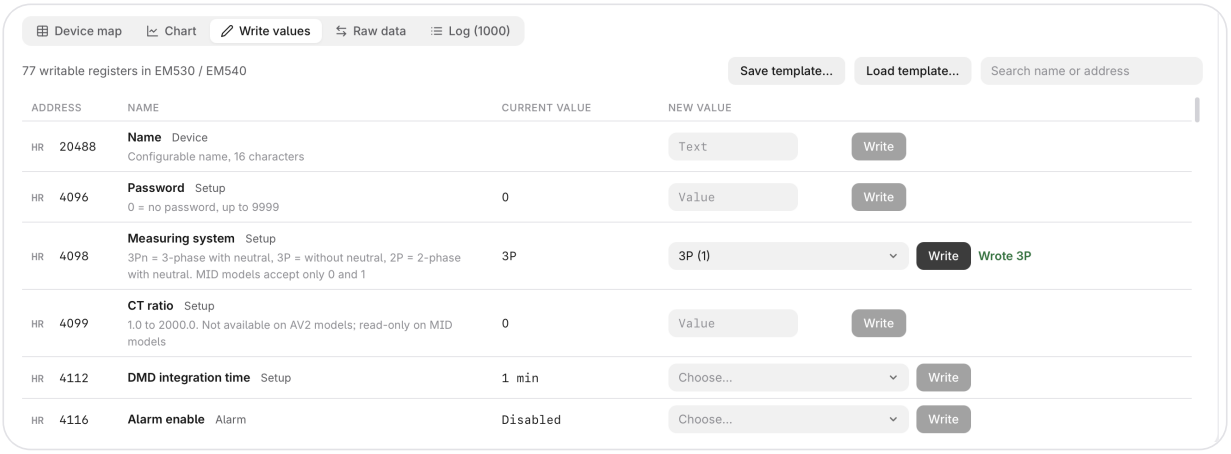


Figure 10. Writable registers of a map; one mode just written.

Limits
A value outside the map’s limits is refused before anything is sent, with the limit it broke.

Bit fields
Writing a bit or a field reads the register first and changes only its bits.

Custom values
For a register with labels, a value that is not one of them can still be typed.

Templates

A template copies the settings of one device to another of the same model. Save template... saves the values last read from the writable registers; Load template... writes the values of a saved template to the device shown.

5.3 Writing safely

Writes change how a real machine behaves. Modbench has three guards, in Settings › Connection:

Confirm before writing to a device
Asks before every write: write commands, map values and templates.

Read-only mode
Blocks every write, also on a connection that is already open. Step 3 shows a lock while it is on; click it to go to the setting.

Map mismatch
After a device does not match its map, writes always ask first.

> Careful
> A write is sent as soon as it is confirmed. Check the unit ID: on a serial line, unit ID `0` is a broadcast that every device takes.

CHAPTER 6

Charts

The `Chart` tab plots values over time while a test runs: map registers by name, or addresses of a manual read.

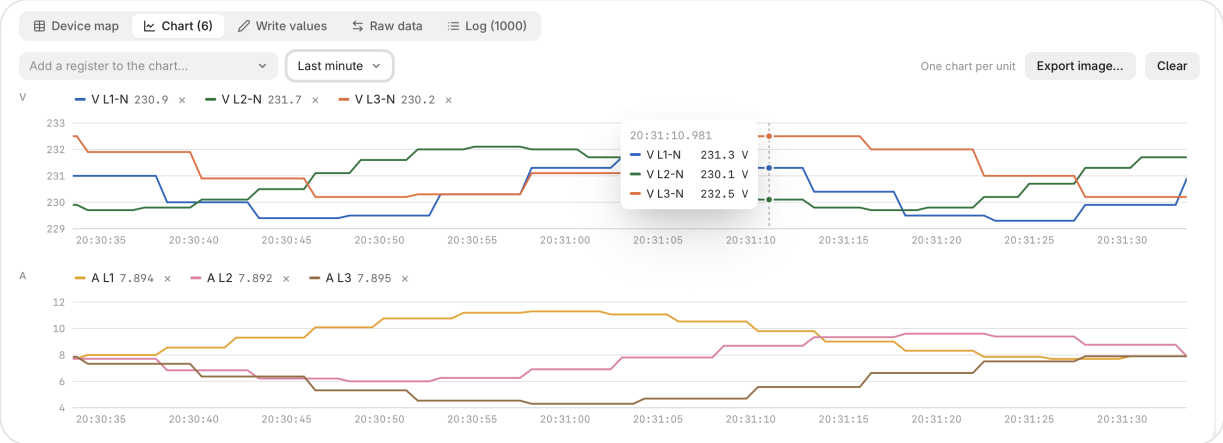


Figure 11. Voltages and currents of a meter, one chart per unit.

6.1 Adding lines

- With a map: **Add a register to the chart** lists the map's numeric registers.
- With a manual read: add an address from the list, or right-click a cell on the `Read registers` tab and choose **Add to chart**.

Up to eight lines can be shown, each in its own colour. Lines with the same unit share a chart, so values of different sizes (volts and amperes) do not flatten each other. The tab's name counts the lines: `Chart (3)`.

Values are recorded from the moment a line is added, while the test runs. To remove a line, click `×` next to its name in the legend.

6.2 Reading the chart

Time span
Time span shows the last minute, 5 or 15 minutes, the last hour, or everything since the lines were added.

Legend
Each line’s name and its newest value.

Hover
Move the pointer over a chart to see every line’s value at that moment.

Spikes
When there are more samples than pixels, each pixel keeps its lowest and highest value, so short spikes stay visible.

Gaps
A failed read leaves a gap in the line instead of a false value.

`Clear` empties the charts and keeps the lines.

6.3 Saving a chart as an image

`Export image...` saves the charts shown, for a report or a bug report:

Choice	What you get
PNG...	A picture at twice the screen size, sharp in documents.
SVG...	A vector drawing that scales to any size and opens in other programs (colours written out, no styles needed).
PNG, light theme...	As PNG, in the light theme even when Modbench is dark.
SVG, light theme...	As SVG, in the light theme.

The image holds every chart shown, each with its unit and legend, under a line with the date and the time span. The hover marks are left out. The image is taken when you choose, so it does not move on while the save dialog is open.

The Trend player (Section 10.5) has the same button.

CHAPTER 7

Raw data and the log

When a device answers strangely, the frames tell why. These two tabs show what was really sent and received.

7.1 The Raw data tab

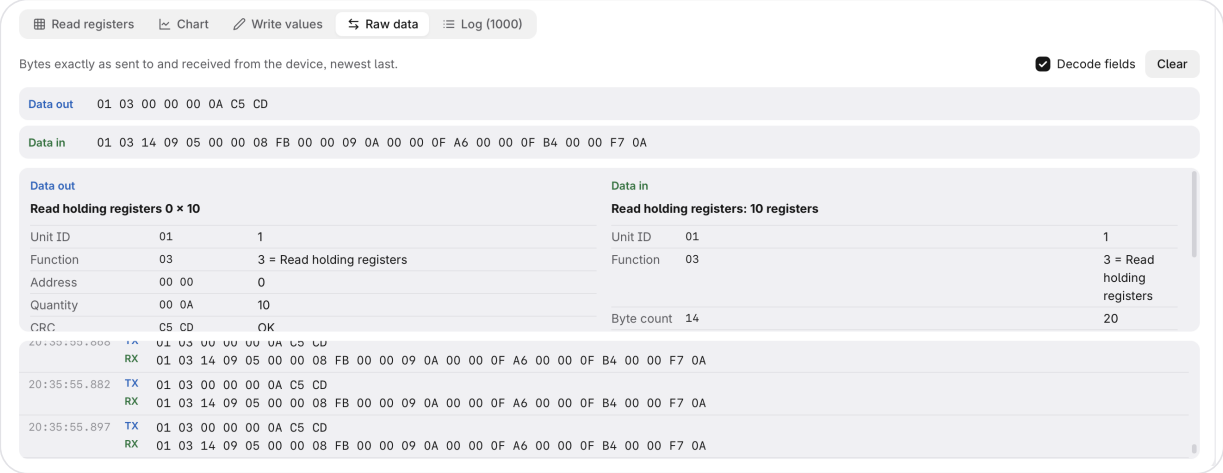


Figure 12. A request and its answer on the Raw data tab.

The bytes exactly as sent to and received from the device, newest last: `Data out` in blue, `Data in` in green, as in the logo. Each frame is decoded field by field (unit ID, function code, address, quantity, byte count, data), and its check is verified: the CRC on RTU, the LRC on ASCII, the header on TCP and UDP. A frame that fails its check is marked.

When no answer came, the tab says so (*No response.*). Clear empties it.

> **Tip**
>
> To read a frame from another program or a bus analyser, paste it into the Frame decoder (Section 10.4): it decodes it the same way.

7.2 The Log tab

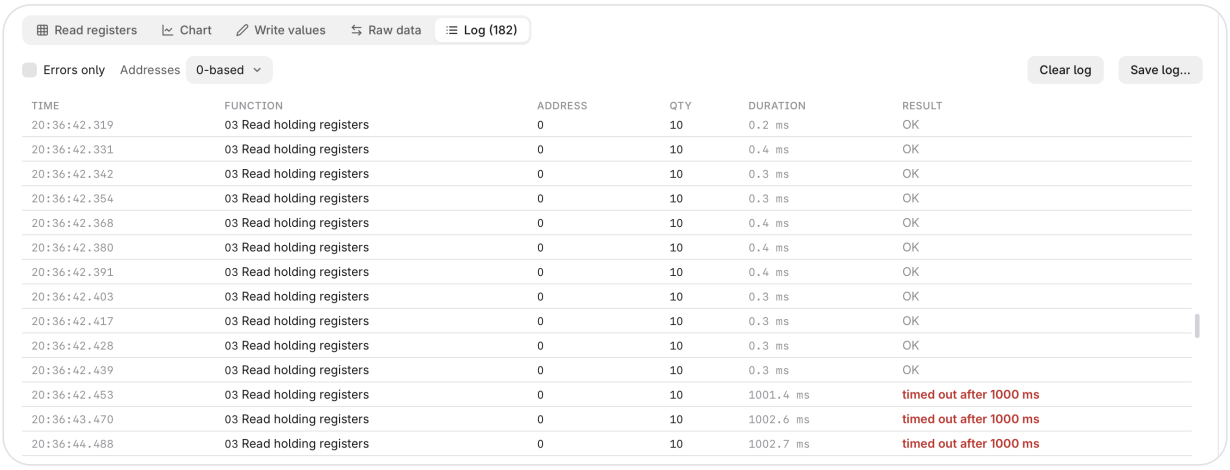


Figure 13. The Log tab, with three timeouts in red.

Every request: **Time**, **Function**, **Address**, **Result** (OK, the exception, or a timeout) and **Duration**. The tab's name counts them: **Log (120)**.

Save log...
Saves the log to a CSV file.

Clear log
Empties it.

Right-click
Copy line, **Copy error**, **Copy all**, and **Show errors only** to keep just the failed requests in view.

Size
Settings › **Connection** › **Requests kept in the log** sets how many requests are kept; older ones are dropped first.

For failures by cause over a long test, see Statistics (Section 10.6).

CHAPTER 8

The device library

The Library page keeps your device maps in one folder, and brings maps from the shared online library.

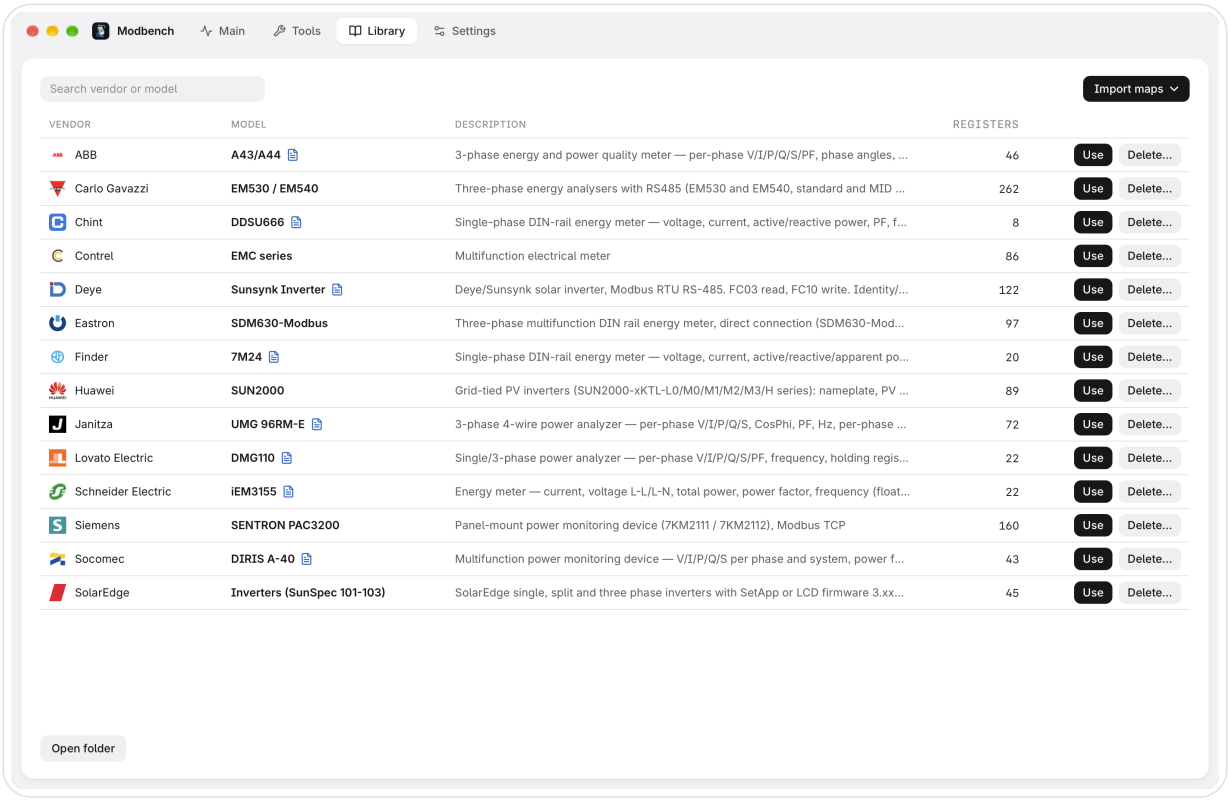


Figure 14. The Library page.

8.1 Your library

The list shows each map’s vendor (with its icon), model, description, number of registers and status (Section 4.4). **Search vendor or model** filters it.

Right-click a map:

Use
Opens it on the Main page.

Simulate this device
Opens the simulator with a device shaped after this map (Chapter 9).

Copy name, Copy file path
For notes and other programs.

Open the library folder
Shows the folder in your file manager.

Delete…
Removes the map from the library, after asking.

Maps with errors are listed with the problem, so they can be fixed in a text editor.

8.2 Adding maps

From a file…
Copies `.mbmap.json` or `.mbmap` files into the library. Version 1 files are converted.

From the online library…
Opens the shared library (below).

The library is a folder (**Settings** › **Device library** › **Library folder**): files copied into it by hand appear too.

8.3 The online library

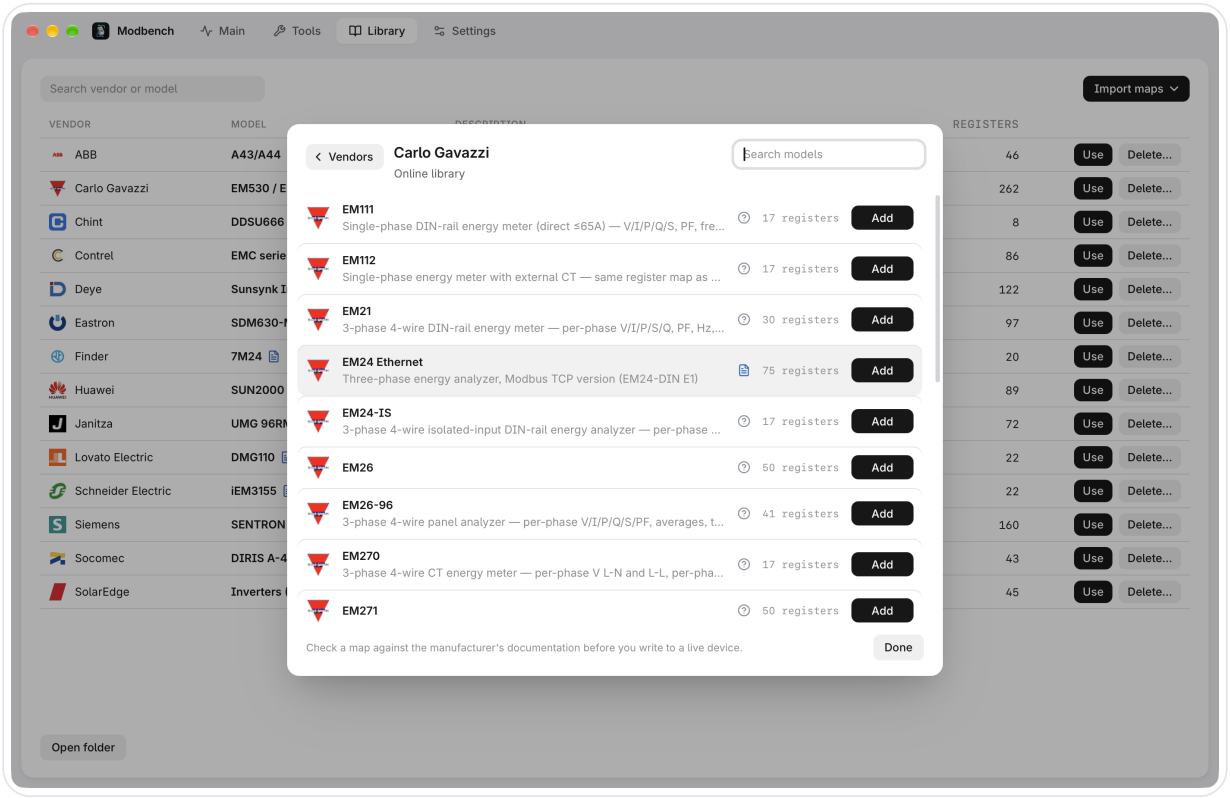


Figure 15. Choosing a map in the online library.

The online library lists the maps shared on GitHub (`github.com/nobrega8/mbmap-library`): over 400 maps of meters, inverters, drives and controllers.

1. Choose a vendor; **Back to all vendors** (or `Esc`) goes back.
2. **Add** next to a model downloads its map into your library.
3. **Done** closes the online library.

A map already in your library can be downloaded again, to replace your copy with the newest one.

> **Share your maps**
>
> A map that works with a real device helps the next person with the same device. The library's repository explains how to contribute one, and the format specification is linked from **Settings** › **Device library**.

CHAPTER 9

The simulator

The simulator runs Modbus devices inside Modbench. Use it to try a map, an alarm or another program (a SCADA, a PLC, a script) without the hardware: each device answers like the real one, with the values of its map, and can make values move on its own.

Open it from **Tools** › **Simulator**.

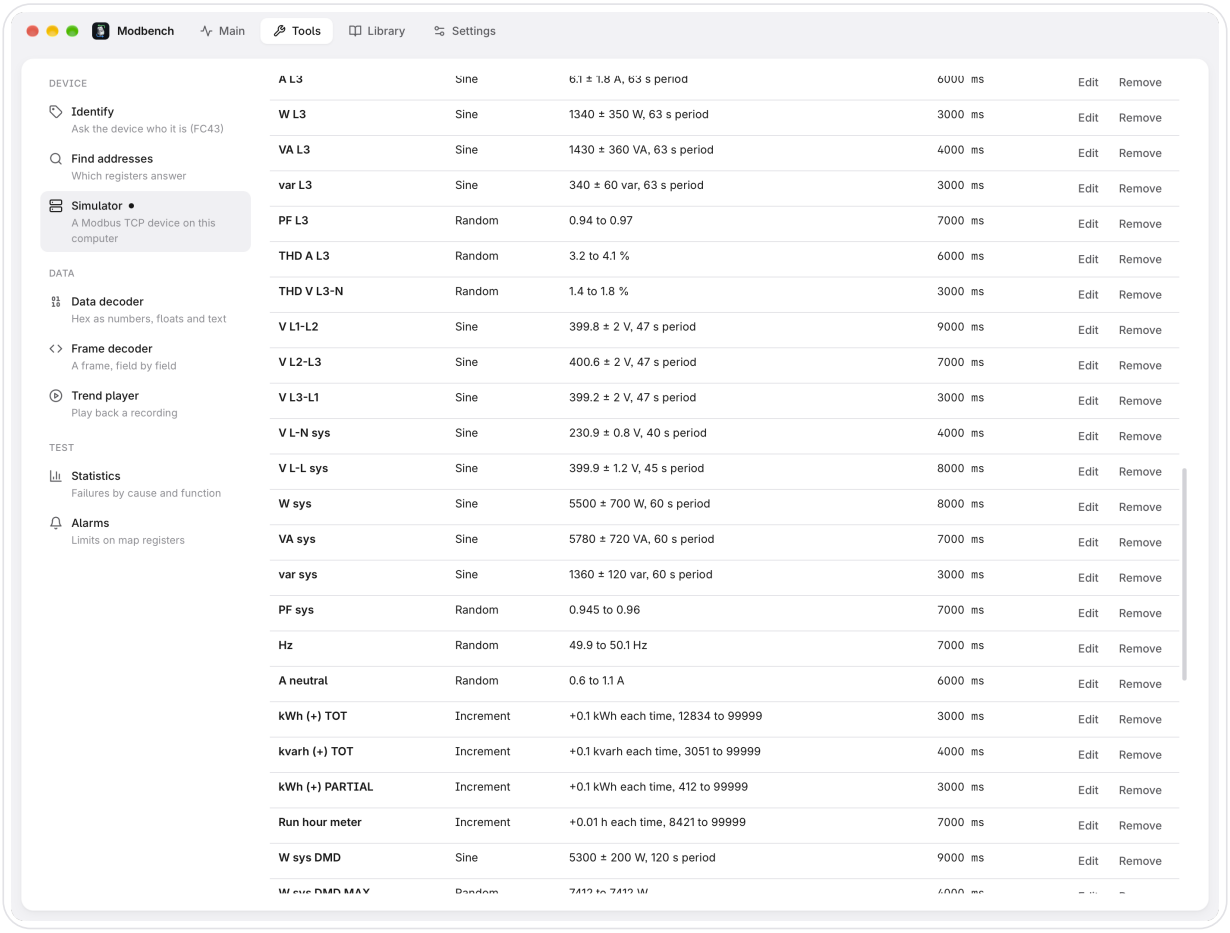


Figure 16. The simulator with two devices on a serial port.

9.1 Where it listens

Interface
Modbus TCP or UDP, RTU over TCP or UDP, or a serial port with RTU or ASCII. Over the network, set the **Port** (5020 by default, so it does not need administrator rights); **Other**

computers also accepts clients from the network, not only from this computer. On a serial port, set the line settings as the clients use them.

Start and **Stop**
Start listens; the status shows where (*Listening on 127.0.0.1:5020*).

Connect Main to it
Fills step 1 of the Main page with the simulator's address and the shown device's unit ID, and goes to Main.

> Serial
>
> Clients connect to the other end of the line. To try it on one computer, use a virtual null-modem pair (com0com on Windows, `socat` on Linux and macOS) and open the other port on the Main page.

9.2 Devices

The simulator starts with one device. Several devices can share one address, each on its own unit ID, like devices behind a gateway or on one RS-485 line; other unit IDs get no answer.

Add device
Adds a device on the lowest free unit ID. The tabs above switch between devices.

Unit ID
The unit ID the device answers. Empty answers every unit ID that no other device has. Unit ID `0` is a broadcast: every device takes the write and none answers.

Map
Shapes the device after a map: one open on the Main page, one from the library, or **Open a map file...**. *Empty device* gives a memory where every address exists and holds zero.

Only the map's addresses
Addresses the map does not have answer exception 2, like the real device. Off, every address answers.

Reset values
Puts back the map's reference values.

Remove device
Removes the device shown; its unit ID gets no answer.

9.3 Values

The view switch shows the device's memory either as **Registers**, the map's registers by name with their values in engineering units, or as a raw **Grid** of addresses, in any area. Click a value to change it: clients read the new value from the next request.

Writes from clients show up immediately, and the recent requests are listed at the bottom, reads and writes apart, so a stream of reads does not push the writes out of sight.

9.4 Generators

Generators make values change on their own, every so many milliseconds (at least 50). Add generator adds one; it drives a map register, in its own units, or a raw address.

Kind	What it does
Increment	Adds a step each time, from a minimum to a maximum, then starts again: counters, energy.
Sine	Swings around an offset with an amplitude and a period: voltage, temperature.
Random	A new value between a minimum and a maximum each time: noise.
Copy	Takes the value of another address: a status input that follows a command coil.

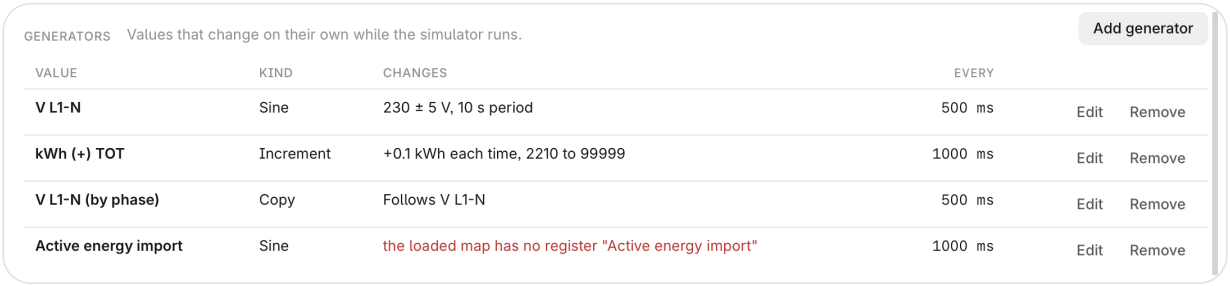


Figure 17. Generators on a meter's registers.

A generator that cannot run (its register is not in the map, or its address does not exist) says why. Generators keep running across Stop and Start.

9.5 Its own window

Pop out moves the simulator into a window of its own, to watch it next to the Main page. Back to the main window brings it back to the Tools page.

9.6 Saving a setup

A test bench with several devices, maps and generators takes a while to set up. Save setup... keeps it in a `.mbsim` file:

- the interface and its settings;
- every device: unit ID, map, *Only the map's addresses*, and generators.

Open setup..., or a double-click on the file, puts it back. Each map is read again from its file when that is still there, so updates to the library are picked up; otherwise the copy kept in the setup is used. While the simulator runs, opening a setup replaces the devices but keeps the interface.

> Note
> Values changed by hand are not saved: devices start from their maps' reference values.

CHAPTER 10

Tools

The Tools page groups the tools by what they work on: the *device*, the *data*, and the *test*. A dot next to a tool means it has something going on (the simulator running, a scan in progress).

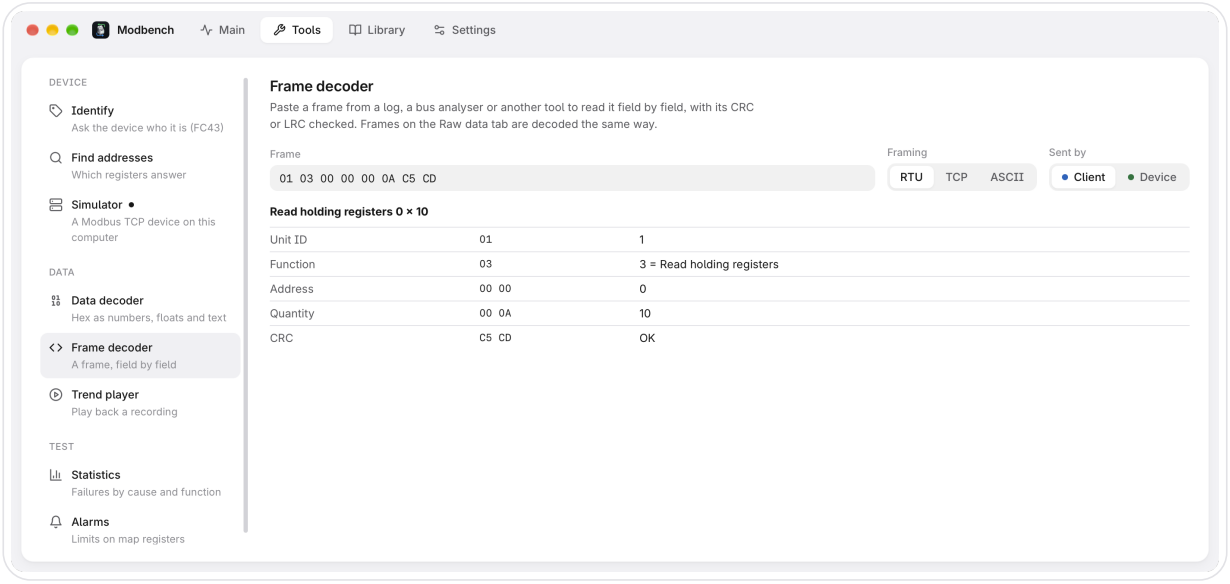


Figure 18. The Tools page.

Tools that talk to a device use the connection of step 1 on the Main page, and only read.

10.1 Identify

Asks the device for its vendor, product code and revision with *Read Device Identification* (FC43/14). Choose which **Objects** to ask for (basic, regular or extended) and Read identification. Each object the device returns is listed with its value.

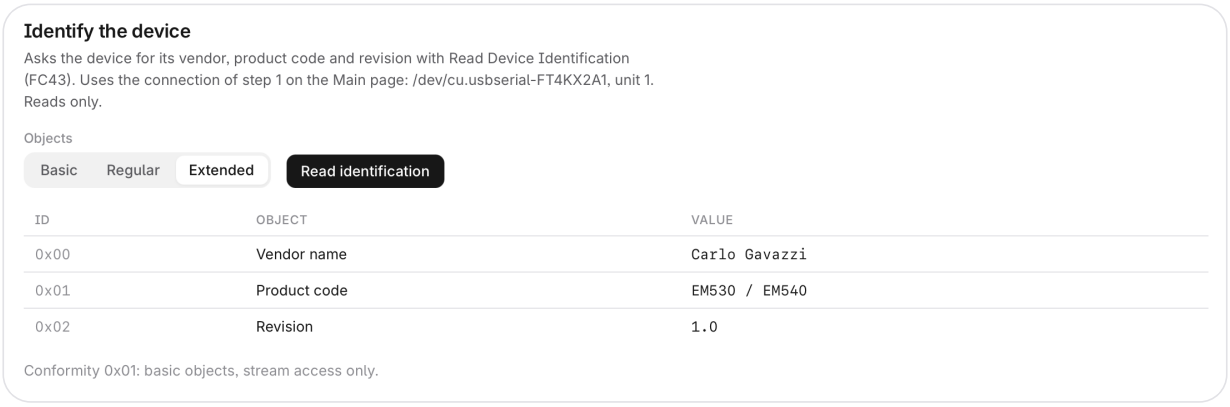


Figure 19. A device that answered FC43.

Not every device supports FC43: many answer exception 1 (illegal function).

Check function codes sends one harmless request for each read function (FC 1–4, 7, 8, 11, 17, 20, 24 and 43) and shows how the device answered: *Supported*; *Not supported* when it answered "illegal function"; *Supported, refused this request* when it answered another exception, which means it has the function but not, for example, address 0; or *No answer*. Nothing is written. FC 7, 8, 11 and 17 exist on serial lines only, so over TCP they often answer "illegal function".

10.2 Find addresses

Finds which addresses answer on a device without a manual. It reads blocks of addresses, and splits the blocks that fail until it knows which addresses exist.

Area, From, To
What to search.

Block
Addresses read at once before splitting. Larger blocks are faster on devices with long runs of registers.

Wait (ms)
How long to wait for each answer.

The result lists the ranges that answered. Read next to a range sets up a manual request for it on the Main page.

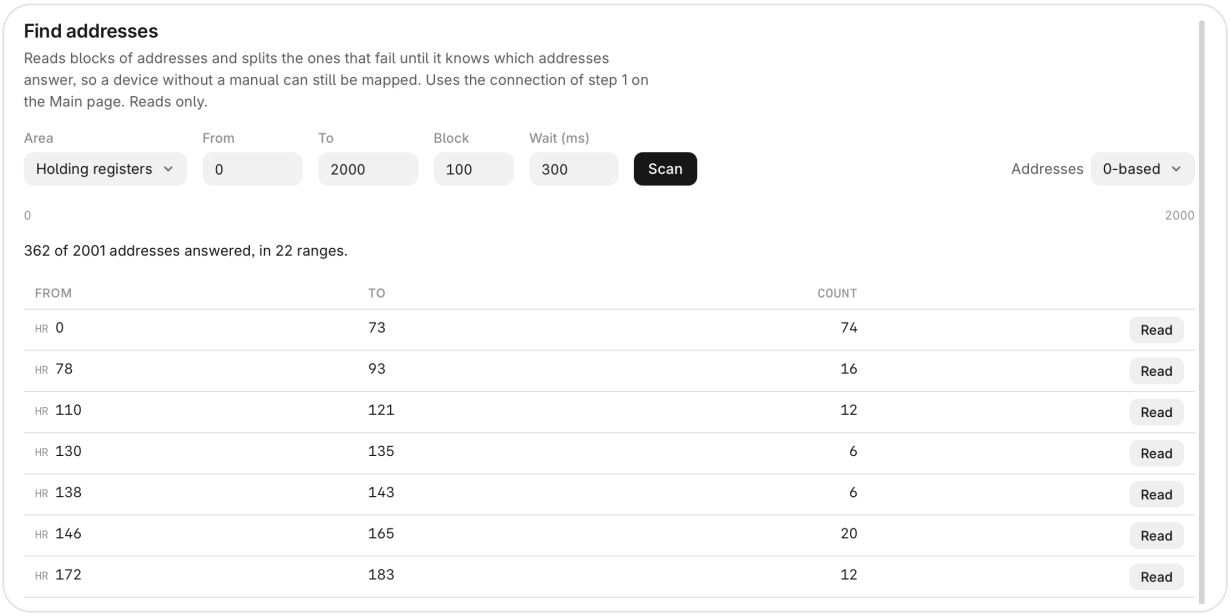


Figure 20. Address ranges found on an energy meter.

10.3 Data decoder

Type or paste registers (in hex) to see them as every type at once, in all four byte orders: signed and unsigned 16-, 32- and 64-bit integers, floats, and text. Use last read takes the registers last read on the Main page.

This is the quickest way to find a value's type and byte order before writing a map: look for the column where the number makes sense (a voltage near 230, a frequency near 50).

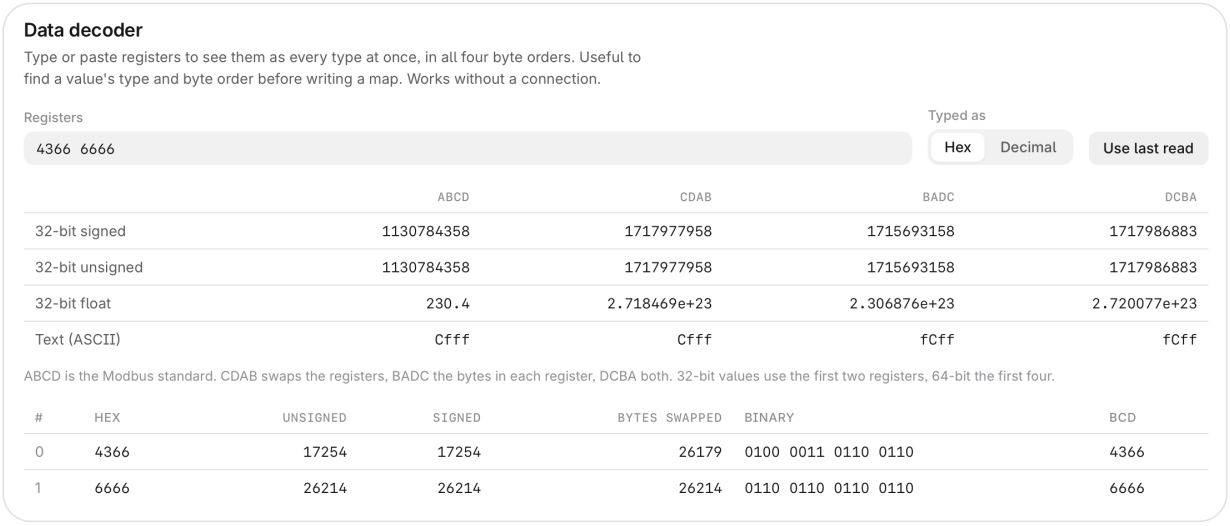


Figure 21. Two registers decoded in every type and order: only 32-bit float in ABCD order reads as a voltage, 230.4.

It works without a connection.

10.4 Frame decoder

Paste a frame from a log, a bus analyser or another tool (for example `01 03 00 00 00 0A C5 CD`) to read it field by field, with its CRC or LRC checked. Choose the **Framing** (RTU, ASCII, TCP) and who sent it (**Client** or **Device**), since a request and an answer with the same function code have different fields.

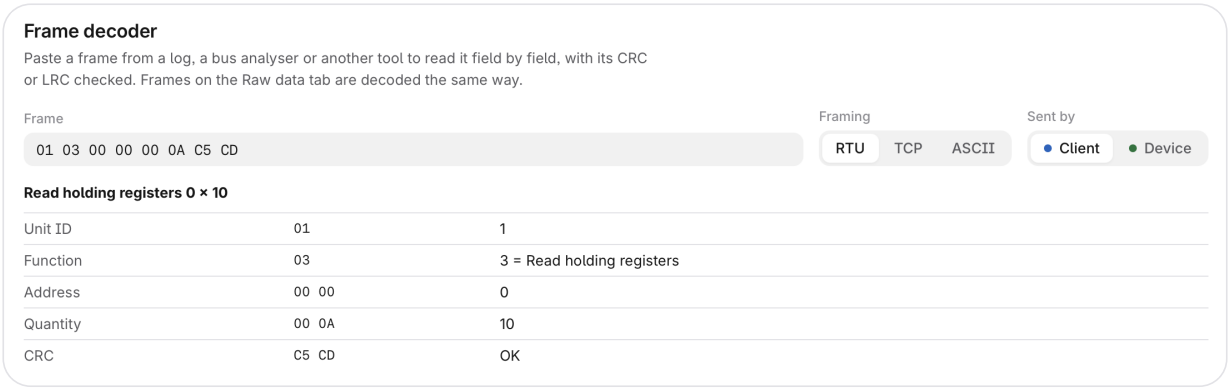


Figure 22. An RTU request decoded.

10.5 Trend player

Plays back a CSV file made with Record... on the Device map tab, so a test can be looked at again later.

1. Open recording... and choose the file.
2. Tick the columns to plot (up to eight; text columns cannot be plotted).
3. Play at 1× to 60× speed, or drag the slider to any moment.

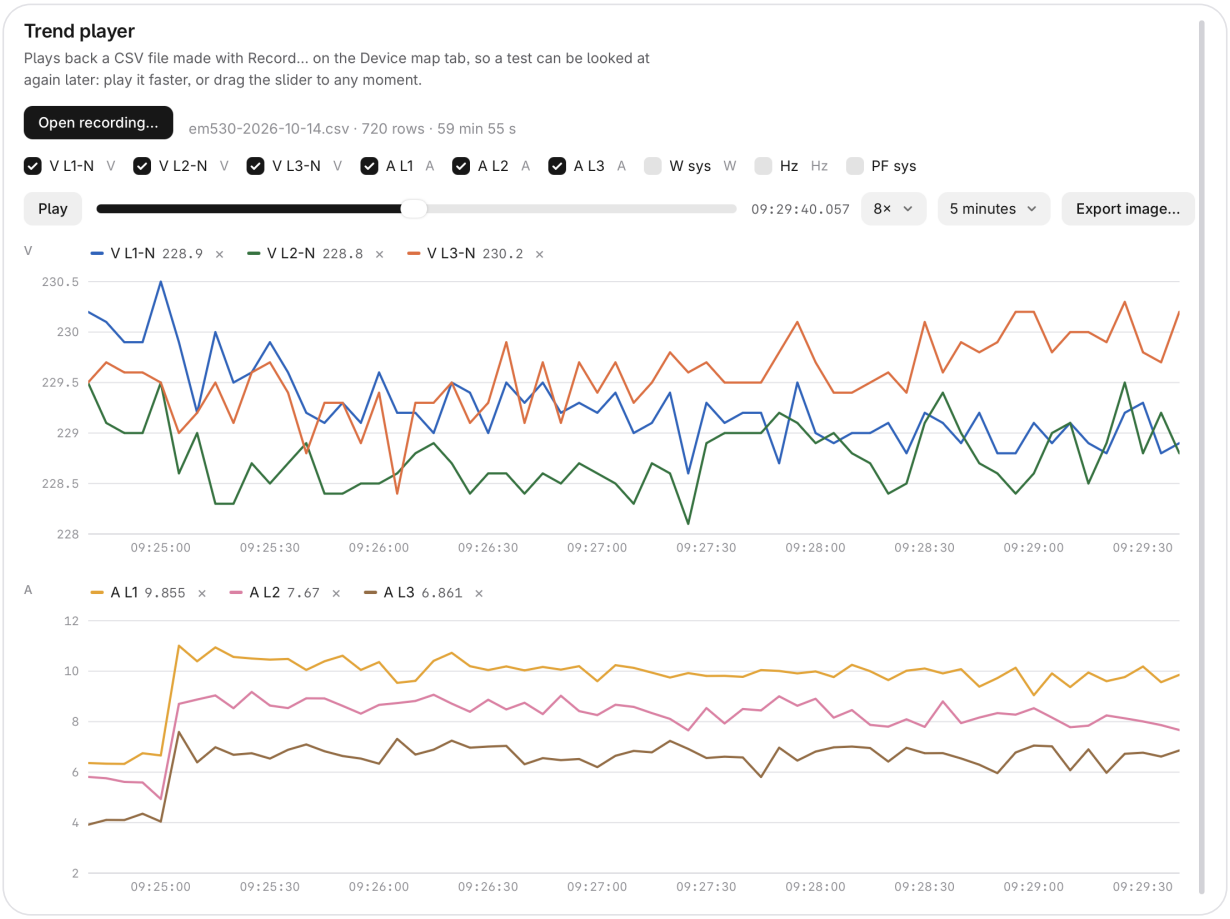


Figure 23. A recording played back.

Time span works as on the Chart tab, and `Export image...` saves the charts (Section 6.3).

10.6 Statistics

Every request sent on the Main page since the last `Reset`:

By result
How many succeeded, timed out, came back as each exception, or arrived as bad frames.
By function
Requests, failures, and minimum, average and maximum response time per function code.

A pattern points to the cause: timeouts to wiring, baud rate or unit ID; exception 2 to an address the device does not have; bad frames to noise or two devices with the same unit ID.

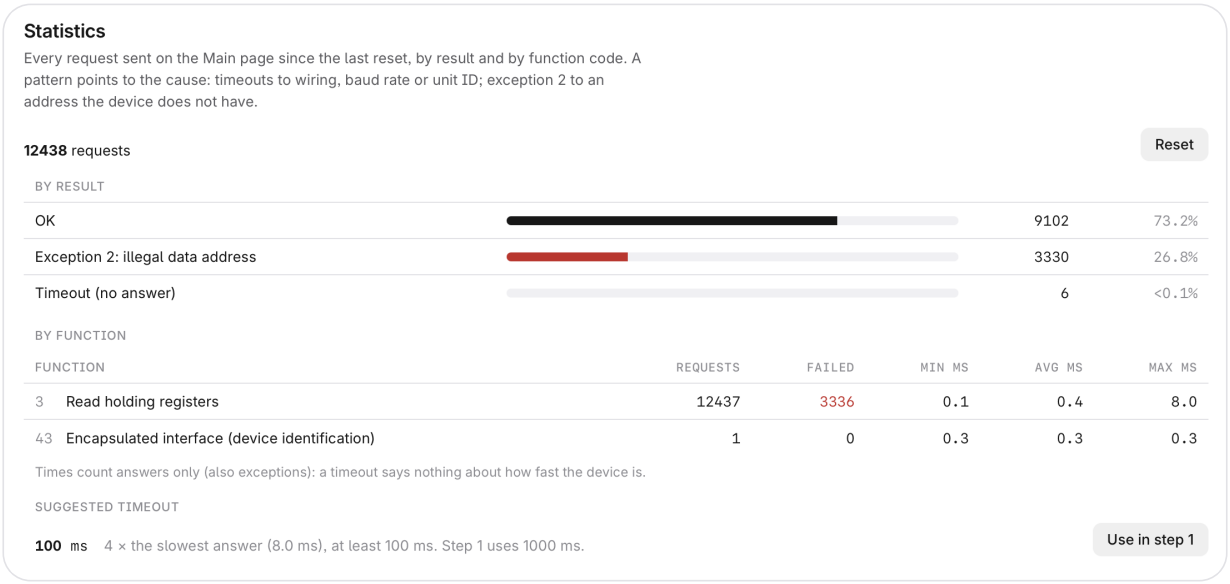


Figure 24. Statistics of a long test.

Suggested timeout is four times the slowest answer; **Use in step 1** puts it in the connection settings.

10.7 Alarms

Limits on the registers of the open map. A value below **Low** or above **High**, or a bit in its alarm state, turns red on the Device map tab.

1. Open a map on the Main page.
2. **Add an alarm on a register...** and choose the register.
3. Set the limits, or for a bit, the state that is an alarm.

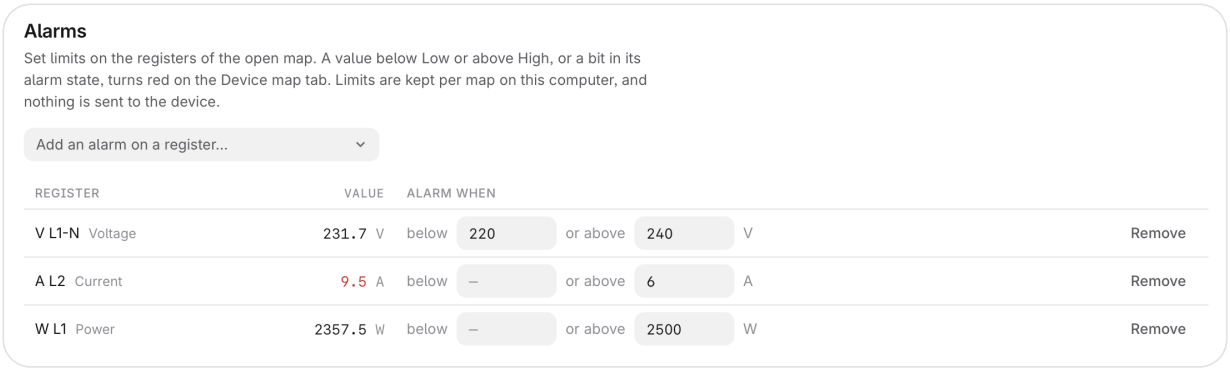
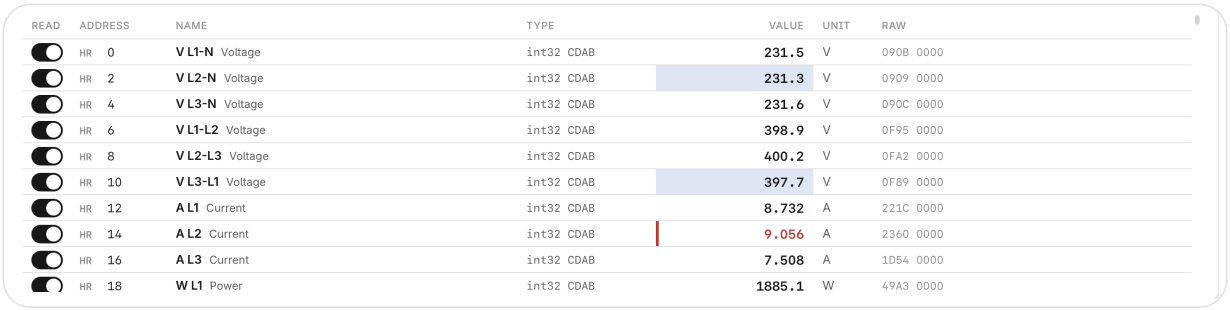


Figure 25. Alarms on a meter’s map.



READ	ADDRESS	NAME	TYPE	VALUE	UNIT	RAW
	HR 0	**V L1-N** Voltage	int32 CDAB	231.5	V	090B 0000
	HR 2	**V L2-N** Voltage	int32 CDAB	231.3	V	0909 0000
	HR 4	**V L3-N** Voltage	int32 CDAB	231.6	V	090C 0000
	HR 6	**V L1-L2** Voltage	int32 CDAB	398.9	V	0F95 0000
	HR 8	**V L2-L3** Voltage	int32 CDAB	400.2	V	0FA2 0000
	HR 10	**V L3-L1** Voltage	int32 CDAB	397.7	V	0F89 0000
	HR 12	**A L1** Current	int32 CDAB	8.732	A	221C 0000
	HR 14	**A L2** Current	int32 CDAB	9.056	A	2360 0000
	HR 16	**A L3** Current	int32 CDAB	7.508	A	1D54 0000
	HR 18	**W L1** Power	int32 CDAB	1885.1	W	49A3 0000

Figure 26. The value in alarm, in red on the Device map tab.

Alarms are kept per map on this computer. Nothing is sent to the device.

CHAPTER 11

Settings

Settings are kept on this computer and apply at once. The list on the left groups them as on the Tools page: *Modbench* (About, Updates), *Display* (Appearance, Values) and *Devices* (Connection, Device library).

11.1 About

The version of Modbench and whether it is up to date, with `Check for updates`. Below it are links to this manual, the release notes, a form to report a problem, the source code and the project's Ko-fi page.

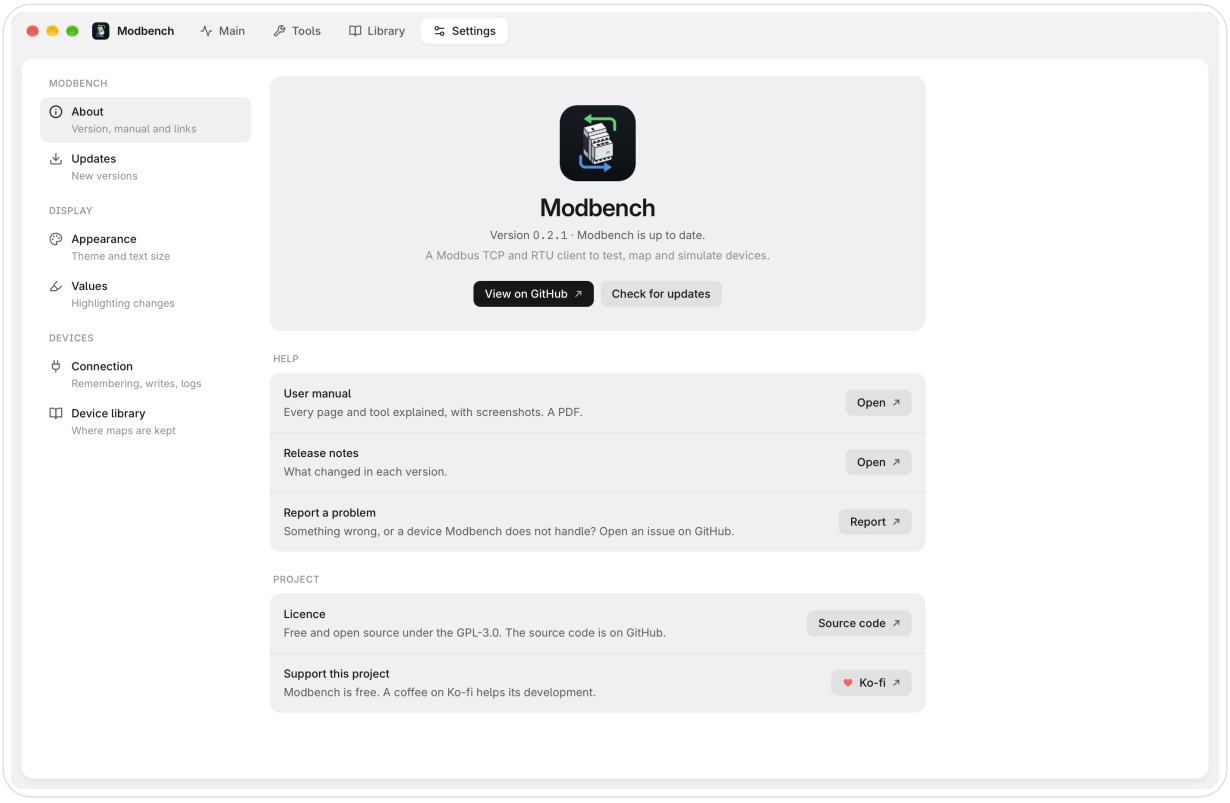


Figure 27. The About section.

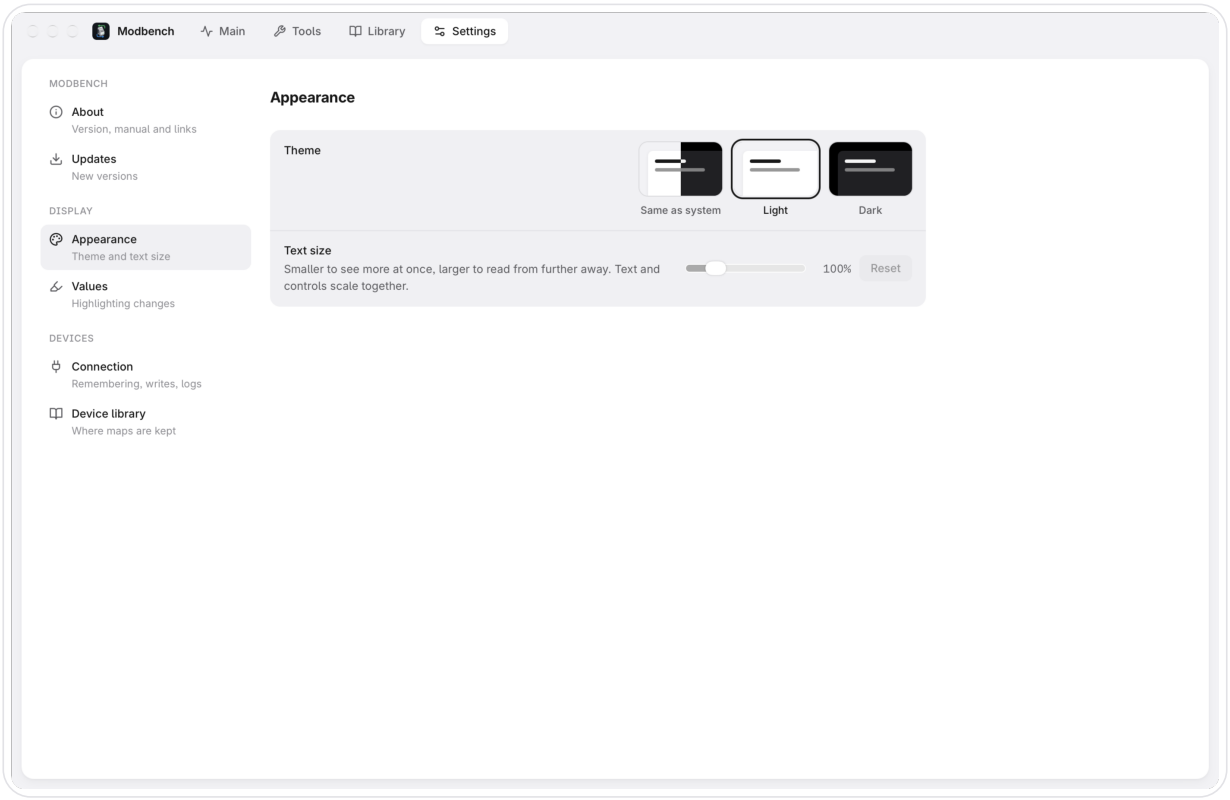


Figure 28. The Settings page, on Appearance.

11.2 Appearance

Theme
Light, dark, or as the system.

Text size
Smaller to see more at once, larger to read from further away. `Reset` goes back to 100 %.

11.3 Values

Highlight changed values
How long a value that changed stays highlighted: off, or 1 to 30 seconds.

11.4 Device library

Library folder
Where your maps are kept. `Open folder` shows it.

Shared maps
Links to the map format and to the repository of maps shared by other users.

11.5 Updates

Check for updates when Modbench starts
On by default. Installs from the APT repository update with `sudo apt upgrade` instead.

11.6 Connection

Remember the last connection
Port, baud rate, address, device and timing are filled in again when Modbench starts.

Remember register choices for each device
Which registers are read, the byte order and Loop come back for each device (map, connection and unit ID). Restoring them never writes or starts a test.

Forget saved device choices
Clears what was remembered per device.

Confirm before writing to a device
Asks before every write.

Reconnect when the connection is lost
While a test runs, a connection that fails is opened again, and the test goes on.

Read-only mode
Blocks every write, also on a connection that is already open.

CSV separator
For saved logs, exported values and recordings. Use a semicolon for spreadsheets set to a language with a decimal comma.

Requests kept in the log
Older requests are dropped first.

11.7 What Modbench keeps

Everything stays on this computer: settings, the last connection, choices per device, alarms, templates and the library folder. The only network requests besides your devices are downloads: the update check, the online library, and the vendors' icons from their websites.

CHAPTER 12

Command line and AI assistants

`modbench-cli` reads devices and works with maps without the window: for scripts, for servers without a screen, and for a quick check. It comes with the app. It never writes to a device.

12.1 Commands

```
modbench-cli read --tcp 192.168.1.10 holding 0x0405 1
modbench-cli read --udp 192.168.1.10 holding 0 10
modbench-cli map --rtu COM3 --unit 18 device.mbmap.json --filter voltage
modbench-cli scan --rtu COM3 --baud 9600 --from 1 --to 32
modbench-cli id --tcp 192.168.1.10
modbench-cli simulate device.mbmap.json --strict
modbench-cli simulate device.mbmap.json --generators moving.json
modbench-cli simulate device.mbmap.json --rtu-over-tcp
modbench-cli simulate device.mbmap.json --rtu COM5 --baud 19200 --parity E
modbench-cli validate device.mbmap.json
modbench-cli convert old-device.mbmap
```

Command	What it does
`read`	Reads coils or registers, like a manual request.
`map`	Reads every register of a map from a device, by name; `--filter` keeps the names that match.
`scan`	Finds the unit IDs that answer.
`id`	Device identification (FC43).
`simulate`	A simulated device shaped after a map (port 5020 by default); `--strict` answers only the map's addresses.
`validate`	Checks a map against the format, with the same messages as the app.
`convert`	Converts a version 1 map to a `.mbmap.json` next to it.

`modbench-cli --help` lists every command and option.

12.2 Generators file

`--generators` takes a JSON list. Each entry names a map register (in engineering units) or a raw `area` and `address`, a `kind`, and how often it changes (`every_ms`, 1000 by default, at least 50):

```
[
  {"register": "Voltage L1", "kind": "sine", "offset": 230, "amplitude": 5, "
    period_s": 10, "every_ms": 500},
  {"area": "input", "address": 0, "kind": "increment", "step": 1, "min": 0, "max":
    100},
  {"area": "holding", "address": 1, "kind": "random", "min": 10, "max": 20},
  {"area": "discrete", "address": 0, "kind": "copy", "from": {"area": "coil", "
    address": 0}}
]
```

12.3 AI assistants (MCP)

`modbench-cli mcp` serves the same reads over the Model Context Protocol (stdio), so an AI assistant such as Claude Code can read devices and check maps: `read_map` reads a device with a map, `validate_map` checks a map file. Like the command line, it never writes to a device.

APPENDIX A

Troubleshooting

A.1 No answer (timeouts)

On a serial line
Check, in this order: the unit ID; baud rate, parity and stop bits (Find... tries them, Section 3.5); the A and B wires, which are often swapped; the termination at both ends of a long line; and that no other master talks on the same line.

On the network
Ping the address; check the port (502, or the converter's); some devices accept one client at a time, so close other programs connected to it. Through a gateway, the unit ID selects the device behind it.

Slow devices
Raise **Timeout**, and add a **Delay** between requests. Statistics suggests a timeout from the measured answers (Section 10.6).

A.2 Values look wrong

Off by one register
The manual counts from 1 and the map, or your request, from 0, or the other way round. Try the address one lower or higher, or check `address_base` in the map.

Huge or tiny numbers
Usually the byte or word order. Try the **Byte order** override in step 2, or paste the registers into the Data decoder (Section 10.3).

Wrong by a factor of 10, 100...
The scale. Maps give it per register; some devices also have a scale factor register.

A mismatch banner
The device is not the one the map is for, or its firmware differs (Chapter 4).

A.3 Bad frames

A CRC or LRC error means the bytes arrived damaged: noise on the line, a missing termination, a wrong baud rate, or two devices answering at once because they share a unit ID. The Raw data tab shows the frame (Section 7.1).

A.4 Exception codes

Code	Name	Usually means
01	Illegal function	The device does not support this function code.
02	Illegal data address	The address, or part of the range, does not exist. Read fewer registers, or check the address base.
03	Illegal data value	A value or a quantity the device refuses.
04	Server device failure	The device failed to do it.
05	Acknowledge	Accepted; the result comes later.
06	Server device busy	Try again later.
08	Memory parity error	The device's memory failed a check.
0A	Gateway path unavailable	The gateway has no route to that unit ID.
0B	Gateway target failed to respond	The device behind the gateway did not answer.

APPENDIX B

Reference

B.1 Function codes

Code	Function	Most per request
01	Read coils	2000 bits
02	Read discrete inputs	2000 bits
03	Read holding registers	125 registers
04	Read input registers	125 registers
05	Write single coil	1 bit
06	Write single register	1 register
15	Write multiple coils	1968 bits
16	Write multiple registers	123 registers
43/14	Read device identification	—

B.2 Byte and word order

How the 32-bit value `0x0A0B0C0D` arrives, in the four orders:

Byte order	Word order	On the wire	Common name
big	big	`0A 0B 0C 0D`	Modbus standard (ABCD)
big	little	`0C 0D 0A 0B`	Word swap (CDAB), used by many meters and PLCs
little	big	`0B 0A 0D 0C`	Byte swap (BADC), rare
little	little	`0D 0C 0B 0A`	Little-endian (DCBA)

B.3 Files

`.mbmap.json`
A device map, version 2: one JSON object with the device's connection defaults and its registers. The format is described in `SPEC.md` of the mbmap-library repository.

`.mbmap`
A device map, version 1 (older, line-based). Modbench opens it and converts it to version 2.

`.mbsim`
A simulator setup (JSON): the interface settings and each device's unit ID, map (with a copy of it), *Only the map's addresses* and generators.

Recordings (`.csv`)
Made with `Record…`: a time column, then one column per register, named with its unit; one row per cycle. The Trend player plays them back.

Logs and exports (`.csv`)
From `Save log…` and `Export…`, with the separator set in Settings.

B.4 Keyboard

Key	Does
`Esc`	Closes a menu or a dialog; in the online library, goes back to the vendors.
`↑` `↓` `Enter`	Move through and choose in menus and lists.
`Tab`	Moves to the next field.